

MAJOR PROJECT REGISTRATION FORM

Date:31-07-2026

Project Category:	☒ Research oriented	☐ Application oriented	Batch ID:	C11
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project:

Foundation Model Guided Physics-Informed Multimodal Diffusion Framework for Cross-Spectral Visible–Thermal Image Translation and Scene Understanding.

Abstract

Thermal imaging has become an essential sensing technology for applications such as intelligent surveillance, autonomous driving, border security, disaster response, and search-and-rescue operations due to its ability to capture heat signatures under low-light, nighttime, and adverse weather conditions. However, thermal images lack the rich color, texture, and semantic details available in visible spectrum images, making accurate human interpretation and high-level computer vision tasks such as object recognition, scene understanding, and identity preservation significantly more challenging. Security agencies, emergency responders, and autonomous systems often rely on thermal imagery in environments where visible cameras perform poorly, yet existing thermal-to-visible image translation methods frequently generate blurred images, lose structural information, and fail to preserve semantic consistency. Recent studies have demonstrated the effectiveness of deep learning approaches; however, they remain limited in maintaining visual realism and cross-modal feature alignment across diverse real-world scenarios.

To address these challenges, this project proposes a **Foundation Model Guided Physics-Informed Multimodal Diffusion Framework for Cross-Spectral Visible–Thermal Image Translation and Scene Understanding**. Unlike conventional CNN- or GAN-based image translation methods, the proposed framework combines foundation model guidance, physics-informed learning, multimodal feature fusion, and diffusion-based image generation to reconstruct realistic visible-spectrum images while preserving structural integrity, semantic relationships, and scene context. The system utilizes publicly available benchmark datasets such as **LLVIP** and **KAIST Multispectral Dataset**, with preprocessing techniques including image normalization, resizing, and data augmentation to improve robustness and generalization. The framework is developed using **Python**, **PyTorch**, **OpenCV**, and supporting scientific computing libraries, while model performance is evaluated using **PSNR**, **SSIM**, **LPIPS**, and **FID** metrics.

The scope of this project is limited to thermal-to-visible image translation and scene understanding using paired multispectral datasets and does not include hardware implementation, real-time embedded deployment, or medical thermal image analysis. The success of the proposed system will be measured by its ability to generate visually realistic visible images with improved structural preservation, perceptual quality, and semantic consistency while outperforming existing baseline models on standard evaluation metrics. The expected outcome is a robust and scalable multimodal framework that enhances cross-spectral image interpretation and supports practical applications in intelligent surveillance, autonomous systems, and emergency response, while providing a strong foundation for future research and publication in advanced computer vision and multimodal artificial intelligence

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References (Category specific)

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